Ex 9a

Implementation of subnetting in CISCO packet Tracer

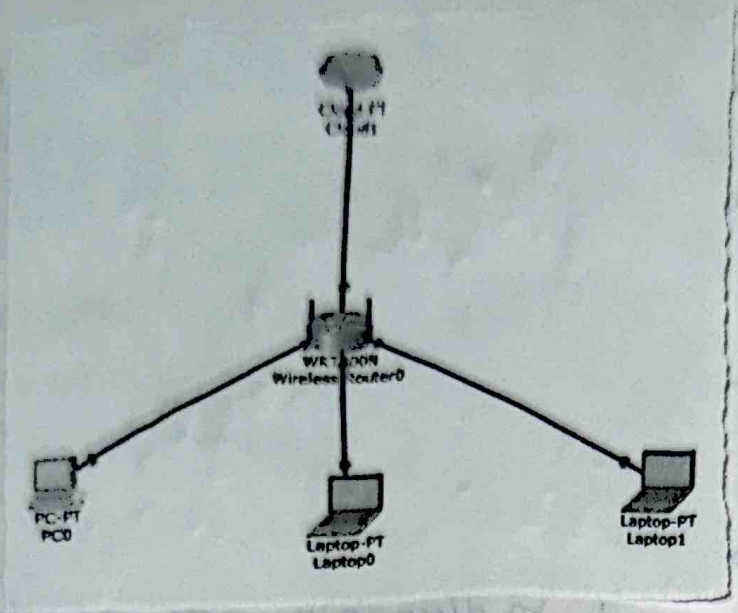
Aim: To implement subnetting by dividing a large network into multiple smaller subnetwork according to required number of host.

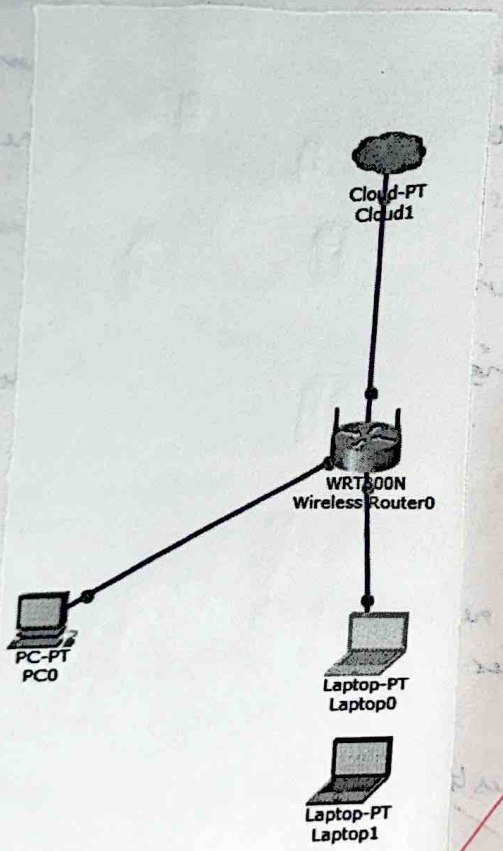
Steps:

1) Open CISCO packet Tracer and create a new project
2) Design the network topology by placing required network device.
3) Select a suitable network device
4) Calculate number of subnet.
5) Determine no of hosts needed
6) Calculate subnet mask
7) Divide the network into multiple
8) Assign a seperate network address
9) verify the subnet calculation
10) Save the network topology.

Result:

Thus multiple subnets were successfully created according to the required number of hosts, and the network address was effiently divided


PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1

Ex: 9b

Assign Static IP Addresses across all subnets and Connect the subnets through a router

Aim :

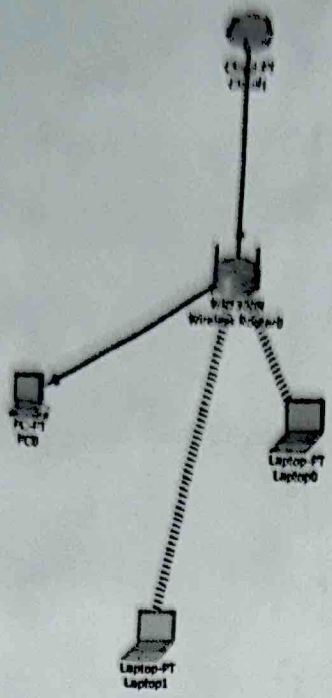
To assign static IP addresses to all devices in different subnets and connect the subnets through a router.

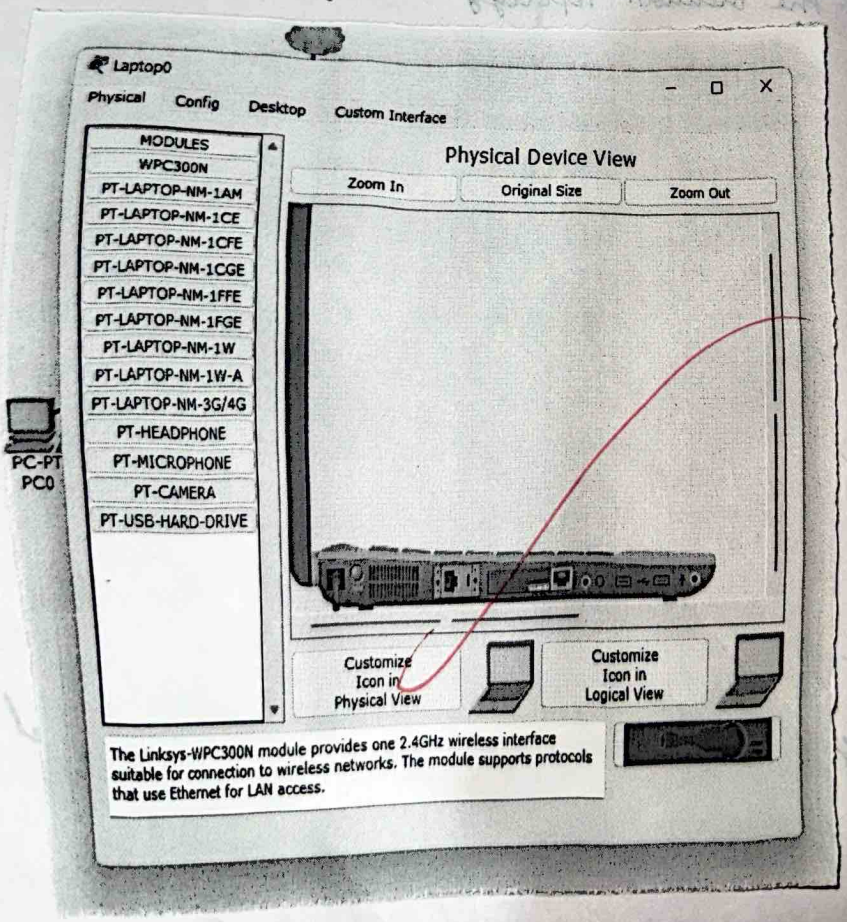
Steps :

1. Place a router in the network topology
2. Connect the router to the switches using appropriate cables
3. Configure the router interface
4. Assign an IP address and subnet mask to each router interface
5. Configure the switches
6. Assign static IP addresses to all computers in every subnet
7. Configure the subnet mask for each device
8. Assign the appropriate default gateway to each computer
9. Verify the IP configuration of all devices
10. Save the router configuration
11. Check the network topology for configuration errors
12. Verify that all devices are properly connected

Result :

Thus, static IP addresses were successfully assigned to all devices, and communication between different subnets was established through the router.


PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1


Laptop0
Physical
Config
Desktop
Custom Interface
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-3G/4G
PT-HEADPHONE
PT-MICROPHONE
PT-CAMERA
PT-USB-HARD-DRIVE
Physical Device View
Zoom In
Original Size
Zoom Out
Customize Icon in Physical View
Customize Icon in Logical View
The Linksys-WPC300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.
PC-PT
PC0

Exp 9c

Simulate Packet Transmissions between the Subnets and Observe the Results

Aim:

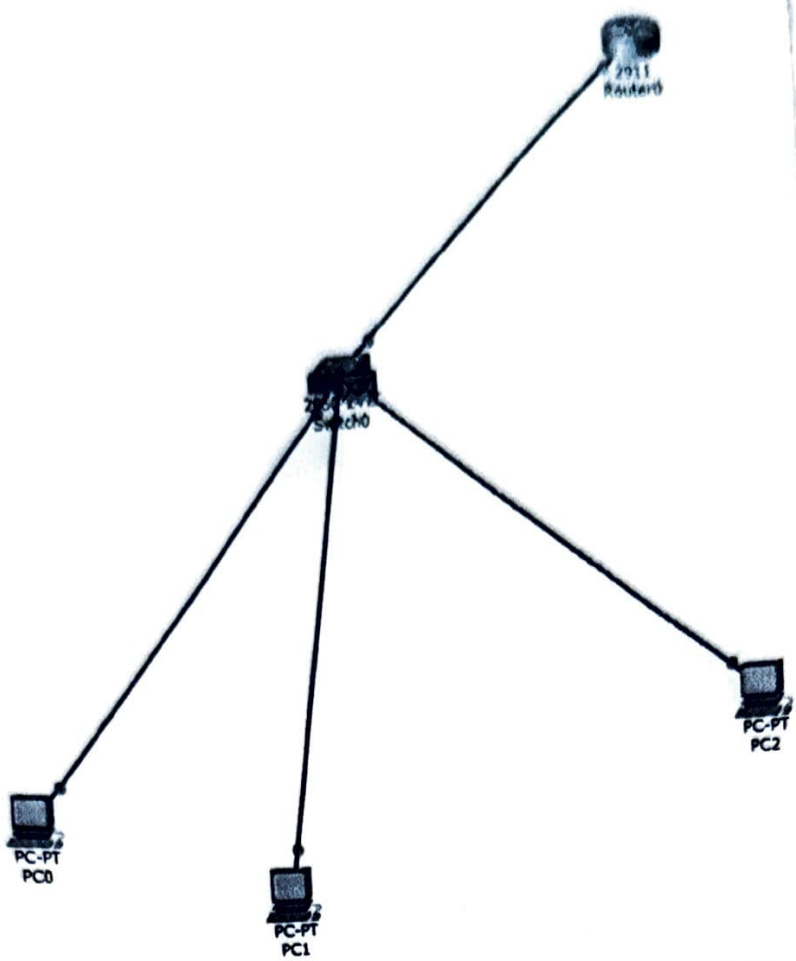
TO analyze packet transmission between different subnets when a router is used for communication.

Steps:

1. open Simulation Mode in Cisco Packet
2. Select a source device from one subnet.
3. Select a destination device from another subnet.
4. Generate a packet using the simple PDU tool.
5. Observe the movement of the pack through the network
6. Analyze how the router receives the packet.
7. Create the forwarding process performed by the router
8. Verify the packet transmission between different subnets
9. Use the ping command to test connectivity
10. Observe the successfull delivery of packets

Result -

This Packets were succfully transmitted between different Subnets through the routes. The routes received the packets from the source subnets. Processed them and forwarded them to the subnet succfull ping




```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no

Press RETURN to get started!


Router>enable
Router$configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)$interface gigabitethernet 0/0
Router(config-if)$ip address 192.168.10.1 255.255.255.0
Router(config-if)$no shutdown

Router(config-if)$
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
```

ii) When subnets are not connected through a router

Aim:

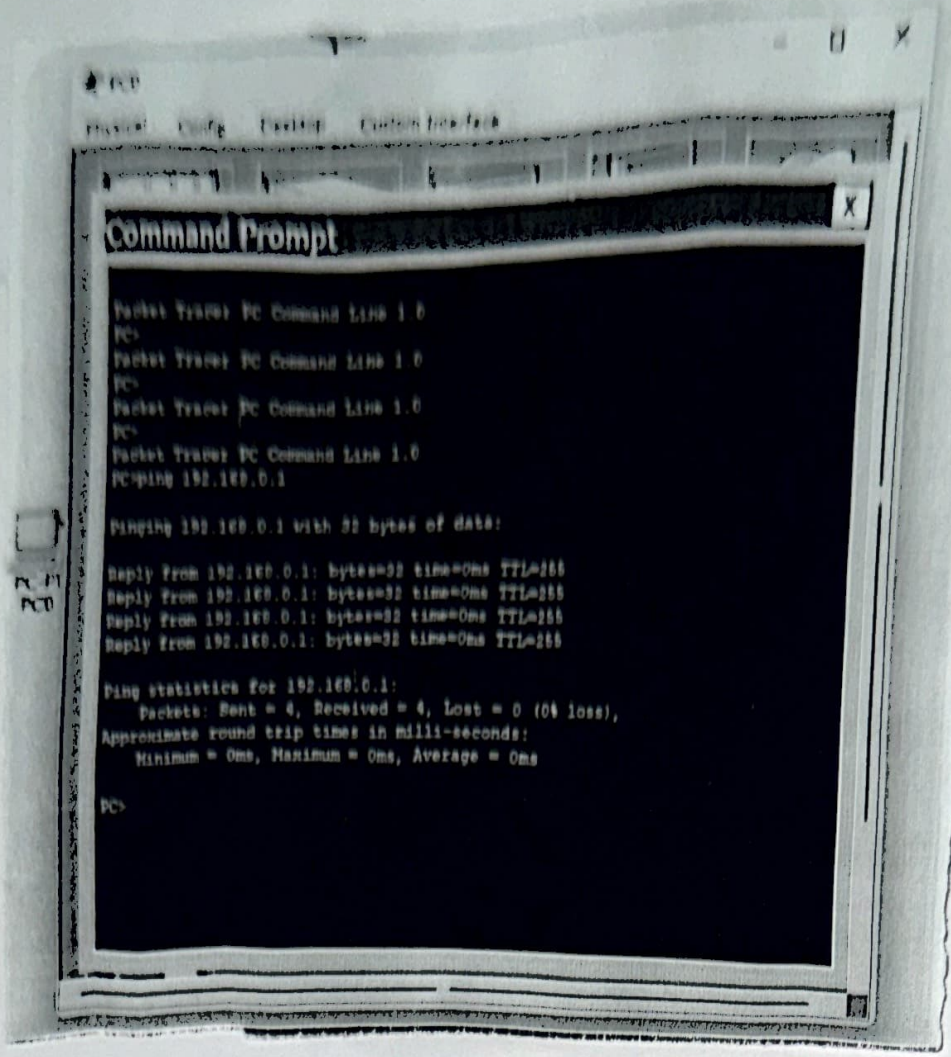
To analyze packet transmission between different subnets when no router is available

Steps:

1. Remove the routers from the network topology
2. Open simulation mode
3. Select devices belonging to different subnets
4. Generate packets using the simple PDU tool
5. Observe the packet transmission
6. Perform a ping test between devices located in different subnets.
7. Analyze the packet flows
8. Observe the simulation events
9. Identify the point at which packet transmission fails
10. Record the observations
11) Analyze the importance of a router in Inter-network communication

Result:

Thus, communication between devices located in different subnets failed when a router was not used. The packets could not reach the destination because there was no routing device to guide them.


Command Prompt
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>